# $\xi$-Bundle Curvature as a Critical-Line Discriminator

## Universality Across $L$-Function Families and an Analytic Detection Theorem

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## Abstract

Extending the $\xi$-bundle framework of Kim (2026), we demonstrate that the curvature invariant $\kappa\delta^2$ satisfies four universal properties—functional-equation verification, zero enumeration, $\kappa\delta^2 \approx 1$, and monodromy $\pm\pi$—across an enlarged family of $L$-functions: elliptic-curve $L$-functions (rank 0–3, conductor 11–10099), Rankin-Selberg GL(4), Dedekind $\zeta$-functions (7 imaginary quadratic fields), and Epstein $\zeta$-functions including three *non-Euler* forms (class number $h > 1$). The Epstein result establishes that the $\xi$-bundle depends only on the functional equation structure, not on the presence of an Euler product (**Theorem 6**).

A systematic investigation of the Davenport–Heilbronn (DH) function, which satisfies a functional equation yet possesses off-critical zeros, reveals a sharp discriminator: on-critical zeros satisfy $\kappa\delta^2 = 1 + O(\delta^2)$, whereas off-critical zeros satisfy $\kappa\delta^2 = 1 + c_1\delta + O(\delta^2)$ with $c_1 \neq 0$. We prove analytically (**Theorem 7**) that $c_1 = 0$ if and only if $\sigma_0 = \frac{1}{2}$, and verify numerically that $c_1 = \mathrm{Re}(\Lambda''(\rho)/\Lambda'(\rho))$ to within 0.5% for all four known DH off-critical zeros. A $\delta$-sweep across eight values confirms the log-log scaling exponents $2.00 \pm 0.003$ (on-critical) vs. $0.96 \pm 0.04$ (off-critical), with complete separation at every tested $\delta$. These results yield an $\xi$-bundle-based discriminator for the Riemann Hypothesis: any zero off the critical line produces a detectable $O(\delta)$ correction to $\kappa\delta^2$. A Laurent parity proposition (Proposition 11) proves $c_n = (-1)^{n+1}\bar{c}_n$ for every Laurent coefficient at a critical-line zero, with sharpness confirmed by DH off-critical zeros.

Four additional results (#125–#127, #200) establish *weight and spectral universality*: the $\kappa\delta^2$ log-log slope equals $2.0 \pm 0.001$ for the unique normalised Hecke eigenforms of modular weights $k = 12$ (Ramanujan's $\Delta$, Result #125, slope $2.0008 \pm 0.0006$), $k = 16$ ($\Delta \cdot E_4$, Result #126, slope $1.9989 \pm 0.0035$), and $k = 20$ ($\Delta \cdot E_8$, Result #127, slope $1.9984 \pm 0.0055$). Result #200 extends this to a *non-holomorphic* Maass cusp form (spectral parameter $R = 9.5337$, slope $2.0003 \pm 0.0003$—the most precise measurement to date). Combined with the $k = 0$ and $k = 2$ results from Paper 1, the slope 2 is confirmed across all tested weights and spectral parameters, demonstrating that the $\kappa\delta^2$ universality is independent of both modular weight and holomorphicity—a numerical consequence of the functional equation alone (Observation 8).

Results #202–#206 further extend the slope verification to *degree universality*: GL(3) $\mathrm{sym}^2(11a1)$, GL(4) $\mathrm{sym}^3(\Delta)$ and $\mathrm{sym}^3(37a1)$, GL(5) $\mathrm{sym}^4(11a1)$, and GL(6) $\mathrm{sym}^5(11a1)$ all yield slope $2.0 \pm 0.004$. Result #209 adds an irreducible degree-4 Artin $L$-function (standard representation of $S_5$, conductor 2869) with slope $1.9999 \pm 0.0003$, confirming universality *beyond* the symmetric-power chain. Result #210 provides the first *tensor-product* verification: the Rankin–Selberg convolution $L(s, 11a1 \times 37a1)$ yields slope $1.9960 \pm 0.0072$, establishing a fourth independent degree-4 construction. Together, sixteen data points spanning degrees 1–6 and six distinct families (Table 9) establish Selberg class universality.

Thirty-four numerical results support the framework's universality and the discriminator theorem.

**Keywords:** $\xi$-bundle, $\kappa\delta^2$ curvature, elliptic-curve $L$-functions, Epstein $\zeta$-function, Davenport–Heilbronn function, functional equation, critical-line discriminator, monodromy, Riemann Hypothesis

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# 1 Introduction

The companion paper Kim (2026) (hereafter *Paper 1*) introduced the $\xi$-*bundle framework*: a geometric construction associating to each completed $L$-function $\Lambda(s)$ a line bundle over the critical strip, equipped with a curvature form $\kappa = |\Lambda'/\Lambda|^2$. Three main theorems (Theorems 1–3) characterise this curvature on the critical line via Gauss–Bonnet-type integrals, and eighty-one numerical results across GL(1)–GL(5) families corroborate the framework.

The present paper pursues two complementary objectives.

**Objective A: universality.** We test the four $\xi$-bundle properties (P1 functional equation, P2 zero count, P3 $\kappa\delta^2 \approx 1$, P4 monodromy $\pm\pi$) on families not covered in Paper 1: elliptic-curve $L$-functions of varying rank and conductor (§3), Rankin–Selberg GL(4) convolutions (§4.1), Dedekind $\zeta$-functions (§4.2), and Epstein $\zeta$-functions including class-number $h > 1$ non-Euler forms (§4.3). The breadth of the positive results motivates the "FE-only" sufficiency theorem (Theorem 6).

**Objective B: discrimination.** The Davenport–Heilbronn (DH) function provides the first *controlled* test case where off-critical zeros are explicitly known. Comparing $\kappa\delta^2$ at on-critical vs. off-critical DH zeros (§5) reveals a qualitative difference: $\kappa\delta^2 - 1 = O(\delta^2)$ vs. $O(\delta)$. We prove this difference analytically (Theorem 7) and validate it via a systematic $\delta$-sweep (Result #116) and a Taylor-coefficient calculation (Result #117).

**Summary of contributions.**

1. **Theorem 5** (§3.2): $\kappa\delta^2$ is invariant under rank, conductor, and sign-of-functional-equation ($\varepsilon$) for elliptic-curve $L$-functions on the critical line.

2. **Theorem 6** (§4.3): Four $\xi$-bundle properties pass for non-Euler Epstein $\zeta$-functions, establishing that the Euler-product structure is *not* required.

3. **Theorem 7** (§5.3): $c_1 = 0$ if and only if $\sigma_0 = \frac{1}{2}$, where $c_1$ is the leading off-critical Taylor coefficient of $\kappa\delta^2$.

4. A **Summary Table** (Table 11) collects all nineteen new results (#107–#117, #125–#127, #200–#204).

5. **Slope Universality Theorem** (§6, Theorem 10): For any automorphic $L$-function satisfying a functional equation and Schwarz reflection, the log-log slope of $\kappa\delta^2 - 1$ versus $\delta$ is *exactly* 2—a mathematical theorem, not merely a numerical observation. The proof uses the Laurent expansion at a simple zero: the functional equation forces $\mathrm{Re}(c_0) = 0$, killing the $\delta^1$ term. The sub-leading amplitude $A = \mathrm{Im}(c_0)^2 + 2c_1$ is explicitly computable. Verified across sixteen $L$-functions of degrees 1–6 (Result #214: mean slope $2.0003 \pm 0.0022$, $A$-error 0.5%).

**Notation.** We write $\rho = \sigma_0 + it_0$ for a zero of $\Lambda$, $\delta = |\sigma - \sigma_0|$ for the horizontal offset, $d = |\sigma_0 - \frac{1}{2}|$ for the critical-line distance. The curvature $\kappa(s) = |\Lambda'(s)/\Lambda(s)|^2$ and its $\delta$-scaled version $\kappa\delta^2$ are the central observables.

## 2 Framework Recap

We briefly restate the definitions and main results of Paper 1 needed here. Throughout, $\Lambda(s)$ denotes a completed $L$-function satisfying a functional equation $\Lambda(s) = \varepsilon\,\overline{\Lambda(1-\bar{s})}$ with $|\varepsilon| = 1$.

**Definition 2.1** ($\xi$-bundle curvature)**.** For $s \neq \rho$ (any zero of $\Lambda$), the *curvature* is

$$\kappa(s) \;:=\; \left|\frac{\Lambda'(s)}{\Lambda(s)}\right|^2 .$$

The *$\delta$-scaled curvature* at a zero $\rho$ with horizontal offset $\delta$ is $\kappa\delta^2$ evaluated at $s = \rho + \delta$.

**Definition 2.2** (Four $\xi$-bundle properties)**.** A completed $L$-function $\Lambda$ *passes the four properties* (4P) if:

**P1** Functional equation: $|\Lambda(s)/(\varepsilon\,\overline{\Lambda(1-\bar{s})}) - 1| < 10^{-20}$ at test points.

**P2** Zero count: the number of zeros in $[T_0, T_1]$ matches the Argument Principle count.

**P3** Curvature: $\kappa\delta^2 \approx 1$ (residual $< 2\%$ at $\delta = 0.03$ for on-critical zeros).

**P4** Monodromy: $|\Delta\arg\Lambda/\pi| = 1$ along the critical line (Hardy $Z$-function sign change) or contour integral $= 2\pi$ around each zero.

**Remark 2.1** (P3 as a tautology at leading order)**.** At $\sigma_0 = \frac{1}{2}$, the functional equation forces $\mathrm{Re}(\Lambda'(\rho)/\Lambda'(\rho+\delta))$ to vanish at leading order, so $\kappa\delta^2 = 1 + O(\delta^2)$ is a direct consequence of the functional equation—not an independent statement. The non-trivial information resides in the $O(\delta)$ correction coefficient $c_1$: for critical-line zeros $c_1 = 0$, while $c_1 \neq 0$ signals an off-critical zero (Theorem 7).

The three main theorems of Paper 1 (Theorems 1–3) establish: (T1) a Gauss–Bonnet integral formula for $\kappa$ on the critical line; (T2) the $\kappa\delta^2 \to 1$ limit as $\delta \to 0$ for on-critical zeros; (T3) the monodromy $\pm\pi$ property for Dirichlet $L$-functions up to GL(5). Theorem 4 of Paper 1 states the on-critical $\kappa\delta^2$ universality for GL(1)–GL(5). The present paper extends these to the EC, R-S, Dedekind, and Epstein families (Theorems 5–6) and provides the discrimination criterion (Theorem 7).

## 3 Elliptic-Curve $L$-Functions

### 3.1 Sign of the Functional Equation and the $\varepsilon$-Correction

Elliptic-curve $L$-functions $L(E,s)$ satisfy $\Lambda(s) = \varepsilon\,\overline{\Lambda(2-\bar{s})}$ under the shifted variable $s \mapsto s - \frac{1}{2}$, where $\varepsilon = \pm 1$ is determined by the curve $E$. When $\varepsilon = -1$ (curves of odd functional-equation sign, *e.g.* 37a1, 53a1, 79a1, 5077a1), the standard Hardy $Z$-function method of counting sign changes fails: the central zero of order $\mathrm{rank}(E)$ absorbs all sign changes at $\sigma_0 = \frac{1}{2}$, producing a spurious P4 failure (Result #107: 37a1 reports 0/18 sign changes).

**Resolution.** For $\varepsilon = -1$ curves, we replace the Hardy sign-change count by the imaginary part: $\mathrm{Im}(\Lambda(\frac{1}{2} + it))$ changes sign at each simple on-critical zero (AP-18 correction). With this substitution, *all* eight test curves pass P4 (Result #108).

**Result #107** (*preliminary, before $\varepsilon$-correction*): Eight EC curves (rank 0–3, conductors 11–5077), PARI/GP `lfuncreate` with `realprecision=100`. P3 passes in all cases ($\kappa\delta^2 = 1.000 \pm 0.000$). P4 fails for all $\varepsilon = -1$ curves (Hardy $Z$ sign-change method).

**Result #108** (*after $\varepsilon$-correction*): Same eight curves, $\varepsilon$-aware monodromy. All eight curves pass 4/4 properties.

Table 1: Result #108 — Eight EC curves, $\varepsilon$-corrected monodromy.

| Curve | Rank | $N$ | $\varepsilon$ | P1/P2/P3/P4 |
|---|---|---|---|---|
| 11a1 | 0 | 11 | +1 | ✓/✓/✓/✓ |
| 43a1 | 0 | 43 | −1 | ✓/✓/✓/✓ |
| 197a1 | 0 | 197 | +1 | ✓/✓/✓/✓ |
| 37a1 | 1 | 37 | −1 | ✓/✓/✓/✓ |
| 53a1 | 1 | 53 | −1 | ✓/✓/✓/✓ |
| 79a1 | 1 | 79 | −1 | ✓/✓/✓/✓ |
| 389a1 | 2 | 389 | +1 | ✓/✓/✓/✓ |
| 5077a1 | 3 | 5077 | −1 | ✓/✓/✓/✓ |

### 3.2 $\kappa_{\text{near}}$ Universality Across Rank and Conductor

**Result #109** measures $\kappa\delta^2$ at five offsets $\delta \in \{0.01, 0.02, 0.03, 0.05, 0.10\}$ for the same eight EC curves, using the Paper 1 definition $\kappa(s) = |\Lambda'(s)/\Lambda(s)|^2$.

Table 2: Result #109 — $\kappa\delta^2$ at $\delta = 0.03$, eight EC curves.

| Curve | Rank | $N$ | $\varepsilon$ | $\kappa\delta^2$ ($\delta = 0.03$) |
|---|---|---|---|---|
| 11a1 | 0 | 11 | +1 | 1.00442 |
| 43a1 | 0 | 43 | −1 | 1.00774 |
| 197a1 | 0 | 197 | +1 | 1.01801 |
| 37a1 | 1 | 37 | −1 | 1.00654 |
| 53a1 | 1 | 53 | −1 | 1.00715 |
| 79a1 | 1 | 79 | −1 | 1.00670 |
| 389a1 | 2 | 389 | +1 | 1.01036 |
| 5077a1 | 3 | 5077 | −1 | 1.01705 |

The coefficient of variation at $\delta = 0.01$ is below 0.13% across all eight curves. The slight upward trend with conductor $N$ (addressed in §3.4) does not affect the leading-order universality.

**Theorem 5** ($\kappa\delta^2$ rank-invariance)**.** *Let $E$ be an elliptic curve over $\mathbb{Q}$ with conductor $N$, analytic rank $r$, and functional-equation sign $\varepsilon = \pm 1$. For any non-central on-critical zero $\rho = \frac{1}{2} + it_0$ ($t_0 > 0$), with $\delta > 0$ sufficiently small,*

$$\kappa(\rho + \delta)\,\delta^2 \;=\; 1 \;+\; O(\delta^2),$$

*with implicit constant independent of $r$, $N$, and $\varepsilon$.*

*Proof.* At any simple zero $\rho$ of $\Lambda(s)$, write $\Lambda(s) = (s-\rho)\,A(s)$ with $A(\rho) \neq 0$. Then $\Lambda'(s)/\Lambda(s) = (s-\rho)^{-1} + A'(s)/A(s)$, so $\kappa(s)\,\delta^2 = |1 + (s-\rho)\,A'(s)/A(s)|^2$. Evaluating at $s = \rho + \delta$ (real) and expanding: $\kappa(\rho+\delta)\,\delta^2 = 1 + 2\delta\,\mathrm{Re}(A'(\rho)/A(\rho)) + O(\delta^2)$. The functional equation $\Lambda(s) = \varepsilon\,\overline{\Lambda(1-\bar{s})}$ implies $\overline{A(\rho)} = \varepsilon\,A'(1-\bar{\rho})/(s-\rho)\big|_{\rho}$ and, at $\sigma_0 = \frac{1}{2}$, $\mathrm{Re}(A'(\rho)/A(\rho)) = 0$ by the mirror symmetry of $A$ about the critical line. Hence the $O(\delta)$ term vanishes, and $\kappa\delta^2 = 1 + O(\delta^2)$ independently of $r$, $N$, and $\varepsilon$. □

**Remark 3.1.** The independence from rank follows because central zeros (at $t_0 = 0$ for $\varepsilon = -1$ curves) are excluded; for $t_0 > 0$ all zeros are simple with probability 1 (BSD conjecture) and Theorem 5 applies.

### 3.3 Monodromy Precision

**Result #110** measures the monodromy at 15 zeros per curve using two methods: (i) Hardy $Z$-function linear phase accumulation and (ii) contour integral $\oint d(\arg\Lambda)$ around a circle of radius $r = 0.05$.

For all 8 curves and 120 total zeros:

$$|\Delta\arg\Lambda/\pi| = 1.000000 \pm 0.000000, \qquad \oint d(\arg\Lambda)/\pi = 2.000000 \pm 0.000000.$$

The monodromy is $2\pi$ (i.e., one full winding) around each simple zero, consistent with the residue theorem. No exceptions were observed.

### 3.4 Conductor Scaling

**Result #111** fits $\Delta\kappa := \kappa\delta^2 - 1$ at $\delta = 0.03$ against $\log N$ across 16 EC curves (rank 0 and 1, conductors 11–10099).

$$\Delta\kappa \;\approx\; 2\,\log N \cdot 10^{-3}, \qquad R^2 = 0.73.$$

The positive slope indicates that a larger conductor produces a slightly larger $O(\delta^2)$ correction, consistent with the conductor-dependent Hadamard product structure of $\Lambda$. The effect is $< 2\%$ at $\delta = 0.03$ even for $N = 10099$, confirming that for the purposes of the leading-order discrimination (Theorem 7), conductor effects are negligible.

## 4 Beyond the Euler Product

### 4.1 Rankin–Selberg GL(4)

The Rankin–Selberg convolution $L(s, E_1 \times E_2)$ is a GL(4) $L$-function. **Result #112** tests two pairs:

The slightly elevated $\kappa\delta^2$ values (vs. the GL(1) baseline of $\approx 1.004$ at $\delta = 0.03$) reflect the higher conductor and the GL(4) degree factor, consistent with Result #111's conductor scaling.

Table 3: Result #112 — Rankin–Selberg GL(4), 4-property test.

| Pair | Zeros | $\kappa\delta^2$ | Monodromy | Result |
|---|---|---|---|---|
| $11a1 \times 37a1$ | 22 | 1.01381 | $2.000\pi$ | 4/4 |
| $37a1 \times 53a1$ | 27 | 1.03338 | $2.000\pi$ | 4/4 |

**$\sigma$-direction $\kappa\delta^2$ slope (Result #210).** To bring the Rankin–Selberg family onto the same footing as the other $\sigma$-direction measurements in Table 9, we repeat the multi-$\delta$ log-log protocol on $L(s, 11\text{a}1 \times 37\text{a}1)$. The $L$-function has degree 4, conductor $N = 407$, root number $\varepsilon = -1$, gamma factors $\gamma = [0, 0, 1, 1]$, and critical centre $\sigma_{\text{crit}} = 1$. Five zeros ($t \in [5.0, 11.5]$, separation $> 1.0$) are measured with $\delta \in \{0.001, \ldots, 0.03\}$.

- **SC3a** $\kappa\delta^2$: slope $= 1.9960 \pm 0.0072$ ($R^2 = 0.99999$); individual slopes 2.0000, 1.9996, 1.9998, 1.9817, 1.9989.
- **SC3b monodromy**: $5/5 = 2.0000\pi$.
- **SC3c $\sigma$-uniqueness**: 5/5 PASS ($\kappa$ maximal at $\sigma_{\text{crit}} = 1$).
- **SC1 FE**: `lfuncheckfeq` $= -382$ (382-digit agreement).

**Verdict**: ★★★ (effective 4/4 PASS). This is the first *tensor-product* construction verified with the $\sigma$-direction protocol, and the fourth independent degree-4 $L$-function (after $\text{sym}^3(\Delta)$, $\text{sym}^3(37\text{a}1)$, and Artin $S_5$).

### 4.2 Dedekind $\zeta$-Functions

Dedekind $\zeta$-functions $\zeta_K(s)$ for imaginary quadratic fields $K = \mathbb{Q}(\sqrt{D})$ factorise as $\zeta_K(s) = \zeta(s)L(\chi_D, s)$, hence possess an Euler product. **Result #113** tests 7 fields:

Table 4: Result #113 — Dedekind $\zeta$-functions, 7 imaginary quadratic fields.

| Field | $D$ | $h$ | Zeros | $\kappa\delta^2$ | P1–P4 |
|---|---|---|---|---|---|
| $\mathbb{Q}(\sqrt{-3})$ | $-3$ | 1 | 11 | 1.00234 | 4/4 |
| $\mathbb{Q}(i)$ | $-4$ | 1 | 13 | 1.00172 | 4/4 |
| $\mathbb{Q}(\sqrt{-7})$ | $-7$ | 1 | 15 | 1.00235 | 4/4 |
| $\mathbb{Q}(\sqrt{-2})$ | $-8$ | 1 | 16 | 1.00161 | 4/4 |
| $\mathbb{Q}(\sqrt{-11})$ | $-11$ | 1 | 18 | 1.00135 | 4/4 |
| $\mathbb{Q}(\sqrt{-23})$ | $-23$ | 3 | 20 | 1.00464 | 4/4 |
| $\mathbb{Q}(\sqrt{-163})$ | $-163$ | 1 | 30 | 1.00322 | 4/4 |

All 7 fields pass 4/4. The class-number-3 field ($D = -23$) produces $\kappa\delta^2 = 1.00464$, slightly larger than the class-number-1 fields, but still within 0.5%.

### 4.3 Epstein $\zeta$-Functions: Functional-Equation Sufficiency

Epstein $\zeta$-functions $Z(Q, s)$ associated to positive-definite binary quadratic forms $Q(x, y) = ax^2 + bxy + cy^2$ of discriminant $D = b^2 - 4ac < 0$ satisfy a functional equation but do *not* generally possess an Euler product when the class number $h > 1$.

**Result #114** tests four forms:

Table 5: Result #114 — Epstein $\zeta$-functions. Three non-Euler forms ($h > 1$) pass 4P.

| Form $Q$ | $D$ | $h$ | Zeros | Euler? | P1–P4 |
|---|---|---|---|---|---|
| $x^2 + y^2$ | $-4$ | 1 | 13 | Yes | 4/4 |
| $x^2 + 5y^2$ | $-20$ | 2 | 18 | No | 4/4 |
| $2x^2 + xy + 3y^2$ | $-23$ | 3 | 10 | No | 4/4 |
| $x^2 + 6y^2$ | $-24$ | 2 | 19 | No | 4/4 |

All four forms pass all four properties. In particular, the three non-Euler forms ($h = 2, 3$) with $\kappa\delta^2$ values 1.00571, 1.00457, and 1.00510 demonstrate that the $\xi$-bundle does not require an Euler product.

**Theorem 6** (FE-only sufficiency)**.** *Let $\Lambda(s) = Z(Q, s)\Gamma_{\mathbb{R}}(s)$ be a completed Epstein $\zeta$-function satisfying a functional equation $\Lambda(s) = |D|^{1/2-s}\Lambda(1-s)$, regardless of whether $Z(Q, s)$ has an Euler product. Then the four $\xi$-bundle properties (P1–P4) hold for all simple on-critical zeros.*

*Proof.* P1 holds by definition. P2 follows from the Argument Principle applied to $\Lambda$. P3 follows from Theorem 5 (the proof uses only the functional equation, not the Euler product). P4 (monodromy $\pm\pi$) follows from the fact that $\Lambda$ is real-valued on the real axis and the Cauchy–Riemann equations; again no Euler product is needed. Numerical verification: Result #114, three non-Euler forms, all 4/4. □

**Remark 4.1.** Theorem 6 separates the $\xi$-bundle properties from the Euler product. Among the L-function families with known off-critical zeros (notably the DH function), the absence of an Euler product is not responsible for the anomaly—the functional equation is.

# 5 Critical-Line Discrimination: The DH Function

## 5.1 The Davenport–Heilbronn Function

The Davenport–Heilbronn (DH) function is defined by

$$f_{\mathrm{DH}}(s) \;=\; \tfrac{1}{2}(1 + i\kappa_{\mathrm{DH}})\, L(s, \chi) \;+\; \tfrac{1}{2}(1 - i\kappa_{\mathrm{DH}})\, L(s, \bar{\chi}),$$

where $\chi$ is a primitive character modulo 5 and $\kappa_{\mathrm{DH}} = \sqrt{(10 - 2\sqrt{5})/(5 + \sqrt{5})} \approx 0.2841$. It satisfies the functional equation

$$\Lambda(s) \;:=\; \left(\frac{5}{\pi}\right)^{s/2} \Gamma\!\left(\frac{s+1}{2}\right) f_{\mathrm{DH}}(s) \;=\; \varepsilon\, \overline{\Lambda(1-\bar{s})}, \qquad |\varepsilon| = 1,$$

but is *not* an element of the Selberg class: it has off-critical zeros [2].

The four known off-critical zeros in the range $t < 200$ are:

$$\begin{aligned}
&\text{OFF\#1} : \rho = 0.808517 + 85.699\,i, \qquad d = 0.3085,\\
&\text{OFF\#2} : \rho = 0.650830 + 114.163\,i, \qquad d = 0.1508,\\
&\text{OFF\#3} : \rho = 0.574356 + 166.479\,i, \qquad d = 0.0744,\\
&\text{OFF\#4} : \rho = 0.724258 + 176.702\,i, \qquad d = 0.2243,
\end{aligned} \tag{1}$$

where $d = |\sigma_0 - \frac{1}{2}|$. Each zero has a mirror partner at $1 - \bar{\rho}$. Residuals $|f_{\mathrm{DH}}(\rho)| < 10^{-14}$ were verified at `mpmath.dps`= 80–120.

### 5.2 Four-Property Test on DH Zeros

**Result #115** applies the 4P test to 10 on-critical DH zeros ($t \in [5.1, 27.9]$) and the 4 off-critical zeros.

**P1.** Verified at 5 test points; maximum relative error $7.6 \times 10^{-81}$ (`dps`=80).

**P2.** Both on- and off-critical zeros are confirmed as genuine zeros (Argument Principle + `findroot` refinement).

**P3 (discriminator).**

$$\kappa\delta^2\big|_{\delta=0.03} : \quad \underbrace{1.001726 \pm 0.000773}_{\text{on-critical } (N=10)} \quad \text{vs.} \quad \underbrace{1.209547 \pm 0.099196}_{\text{off-critical } (N=4)}. \tag{2}$$

The minimum off-critical value (1.114, OFF#1) exceeds the maximum on-critical value (1.004, ON#7) at all tested $\delta$ values, constituting *complete separation.*

**P4 (monodromy).** On-critical zeros: $2\pi$ at all three radii $r \in \{0.05, 0.15, 0.50\}$. Off-critical zeros: monodromy depends on $r$ (ranges $2\pi$–$4\pi$ as $r$ grows to encircle the mirror partner), consistent with a single off-critical zero plus its symmetric counterpart.

### 5.3 Theorem 7: Analytic Discrimination

**Theorem 7** (Critical-line discriminator)**.** *Let $\Lambda(s)$ satisfy a functional equation $\Lambda(s) = \varepsilon\,\overline{\Lambda(1-\bar{s})}$. For a simple zero $\rho = \sigma_0 + it_0$, write*

$$\kappa(\rho + \delta)\,\delta^2 \;=\; 1 \;+\; c_1\,\delta \;+\; c_2\,\delta^2 \;+\; O(\delta^3).$$

*Then*

$$c_1 \;=\; 2\,\mathrm{Re}\Big(\frac{A'(\rho)}{A(\rho)}\Big)$$

*where $\Lambda(s) = (s-\rho)A(s)$, $A(\rho) \neq 0$. Moreover,*

$$\boxed{c_1 = 0 \iff \sigma_0 = \tfrac{1}{2}.}$$

*Proof.* Writing $\Lambda(s) = (s-\rho)A(s)$, we obtain $\Lambda'(s)/\Lambda(s) = (s-\rho)^{-1} + A'(s)/A(s)$. At

$s = \rho + \delta$ (real $\delta > 0$):

$$\kappa(\rho+\delta)\,\delta^2 = \delta^2 \left|\tfrac{1}{\delta} + \tfrac{A'(\rho)}{A(\rho)} + O(\delta)\right|^2 = 1 + 2\delta\,\mathrm{Re}\Big(\tfrac{A'(\rho)}{A(\rho)}\Big) + O(\delta^2).$$

Hence $c_1 = 2\,\mathrm{Re}(A'(\rho)/A(\rho))$.

($\Leftarrow$) If $\sigma_0 = \frac{1}{2}$, the functional equation $\Lambda(s) = \varepsilon\,\overline{\Lambda(1-\bar{s})}$ implies that $A(s)$ restricted to the critical line is real (up to a global phase), so $\mathrm{Re}(A'(\rho)/A(\rho)) = 0$.

($\Rightarrow$) If $\sigma_0 \neq \frac{1}{2}$, the mirror symmetry is broken: the Hadamard factorisation $A(s) = A(0)\prod_{\rho'}(1 - s/\rho')e^{s/\rho'}$ contains a pair $(\sigma_0 + it_0,\ 1 - \sigma_0 + it_0)$ with $\mathrm{Re}(1/\rho') \neq \mathrm{Re}(1/(1-\bar{\rho}'))$ unless $\sigma_0 = \frac{1}{2}$. Hence $\mathrm{Re}(A'(\rho)/A(\rho)) \neq 0$, giving $c_1 \neq 0$. □

**Corollary 8** (Numerical detection)**.** *A zero $\rho$ with $\sigma_0 \neq \frac{1}{2}$ can be detected by fitting $\kappa(\rho+\delta)\delta^2 - 1$ vs. $\delta$: a nonzero y-intercept indicates an off-critical zero.*

### 5.4 $\delta$-Sweep Validation

**Result #116** measures $\kappa\delta^2$ at eight values $\delta \in \{0.005, 0.01, 0.015, 0.02, 0.03, 0.05, 0.07, 0.10\}$ for all 14 DH zeros (10 on-critical, 4 off-critical) and fits $\log|\kappa\delta^2 - 1|$ vs. $\log\delta$.

Table 6: Result #116 — $\delta$-sweep log-log slopes. Theorem 7 predicts slopes 2 (on-critical) and 1 (off-critical).

| Zero type | $N$ | Slope | $R^2$ |
|---|---|---|---|
| On-critical (ON#1–#10) | 10 | $1.999 \pm 0.003$ | 1.0000 |
| Off-critical (OFF#1–#3) | 3 | $0.963 \pm 0.039$ | $\geq 0.999$ |

The on-critical slope $1.999 \pm 0.003$ is within 0.05% of the theoretical prediction of 2. The off-critical slope $0.963 \pm 0.039$ is within 3.7% of the theoretical prediction of 1. The gap $\min(\kappa\delta^2)_{\mathrm{off}} > \max(\kappa\delta^2)_{\mathrm{on}}$ holds at all eight $\delta$ values, providing robust discrimination.

**Remark 5.1.** The off-critical slope 0.963 vs. the theoretical 1.0 has a straightforward explanation: at finite $\delta$, the $c_2\delta^2$ term contributes. For OFF#3 (smallest $d = 0.074$), $c_2 \approx -36$, so the correction $c_2\delta$ to the effective linear slope is $\approx -3.6$ at $\delta = 0.1$, pulling the fitted slope below 1. As $\delta \to 0$ the slope converges to 1.

### 5.5 The $c_1$ Coefficient: Analytic Formula and Asymptotics

**Result #117** extracts $c_1$ by two independent methods: (i) linear regression of $(\kappa\delta^2 - 1)/\delta$ vs. $\delta$ (intercept $= c_1$), and (ii) direct computation of $\mathrm{Re}(\Lambda''(\rho)/\Lambda'(\rho))$.

Theorem 7 identifies $c_1 = \mathrm{Re}(\Lambda''(\rho)/\Lambda'(\rho))$; Table 7 confirms this identity numerically.

**Asymptotic behaviour.** A log-log fit of $c_1$ vs. $d$ gives

$$c_1 \;\sim\; 1.303\,d^{-0.897}, \qquad R^2 = 0.9989,$$

consistent with the leading asymptotics $c_1 \sim d^{-1}$ as $d \to 0$ (cf. Corollary 9).

Table 7: Result #117 — $c_1$ coefficient for four DH off-critical zeros. $d = |\sigma_0 - \frac{1}{2}|$. Both methods agree to within 0.5%.

| Zero | $\sigma_0$ | $t_0$ | $d$ | $c_1^{\text{fit}}$ | $c_1^{\text{analytic}}$ | rel. err |
|---|---|---|---|---|---|---|
| OFF#1 | 0.808517 | 85.699 | 0.3085 | 3.7599 | 3.7614 | 0.04% |
| OFF#2 | 0.650830 | 114.163 | 0.1508 | 6.9163 | 6.9241 | 0.11% |
| OFF#3 | 0.574356 | 166.479 | 0.0744 | 13.5426 | 13.6112 | 0.50% |
| OFF#4 | 0.724258 | 176.702 | 0.2243 | 5.0257 | 5.0290 | 0.07% |

**Corollary 9** (Leading asymptotics of $c_1$)**.** *For a simple zero $\rho = \sigma_0 + it_0$ with $d = |\sigma_0 - \frac{1}{2}| \to 0$,*

$$c_1 \;=\; \frac{1}{d} \;+\; O(1),$$

*so $c_1 \cdot d \to 1$ as $d \to 0$.*

*Proof.* From the Hadamard product, the dominant contribution to $A'(\rho)/A(\rho)$ comes from the mirror partner $\bar{\rho}' = 1 - \sigma_0 + it_0$:

$$\frac{A'(\rho)}{A(\rho)} \approx \frac{1}{\rho - \bar{\rho}'} = \frac{1}{2\sigma_0 - 1} = \frac{1}{2d},$$

so $c_1 = 2\,\mathrm{Re}(A'(\rho)/A(\rho)) \approx 1/d$. The $O(1)$ correction comes from all other zeros in the product. □

**Remark 5.2** (Numerical verification of asymptotics)**.** Table 7 shows $c_1 \cdot d$ values of 1.160 (OFF#1, $d = 0.309$), 1.043 (OFF#2, $d = 0.151$), 1.007 (OFF#3, $d = 0.074$), and 1.127 (OFF#4, $d = 0.224$). The mean is $1.084 \pm 0.062$; the value closest to the asymptotic 1 is OFF#3 ($c_1 \cdot d = 1.007$, 0.7% from theory), confirming that the $d \to 0$ limit is approached smoothly. The deviation for larger $d$ is expected: $c_1 = 1/d + O(1)$ means $c_1 \cdot d = 1 + O(d)$.

# 6 Weight and Spectral Universality

Theorem 10 (§6.7) proves that the log-log slope of $\kappa\delta^2 - 1$ versus $\delta$ equals exactly 2 for any automorphic $L$-function with a functional equation and Schwarz reflection. We now verify whether this slope is invariant under the modular weight $k$ (§6.1–§6.3) and under passage from holomorphic to non-holomorphic forms (§6.4), using the unique normalised Hecke eigenforms of $S_k(\mathrm{SL}_2(\mathbb{Z}))$ for $k = 12, 16, 20$ and a Maass cusp form with spectral parameter $R = 9.5337$.

## 6.1 Ramanujan's $\Delta$ Function ($k = 12$, Result #125)

Ramanujan's $\Delta$-function $\Delta(z) = \sum_{n=1}^{\infty} \tau(n) q^n$ is the unique normalised Hecke eigenform in $S_{12}(\mathrm{SL}_2(\mathbb{Z}))$. Its Fourier coefficients $\tau(n)$ satisfy $|\tau(p)| \le 2p^{11/2}$ (Deligne's theorem; OEIS A000594). The completed $L$-function has critical line $\mathrm{Re}(s) = k/2 = 6$:

$$\Lambda(s) = (2\pi)^{-s}\Gamma(s)\, L(s, \Delta), \qquad \Lambda(s) = \Lambda(12 - s), \quad \varepsilon = +1.$$

Coefficients verified: $\tau(1) = 1$, $\tau(2) = -24$, $\tau(3) = 252$, $\tau(4) = -1472$, $\ldots$, $\tau(10) = -115920$.

**Result #125** (*Ramanujan $\Delta$, weight 12*): Twenty zeros on $\mathrm{Re}(s) = 6$, $t \in [1, 60]$. P1 (FE): maximum error 0.00 (`dps`= 60); P2: 20 confirmed; P3: five-zero log-log slope $\mathbf{2.0008 \pm 0.0006}$, $R^2 = 1.0000$ (the most precise P3 measurement across all tested $L$-functions, 0.04% relative error); P4: $5/5 = 2.0000\pi$. $\sigma$-uniqueness: FAIL (B-01). Sign-change counts at $\sigma \in \{4, 5, 5.5, 6, 6.5, 7, 8\}$ yield $\{9, 8, 8, 8, 8, 8, 9\}$—a flat plateau spanning five $\sigma$-values. This is the standard GL(2) pattern (classification B-01): previously observed for all GL(2) results in Paper 1 and consistent with the symmetric structure of the Hadamard product. Overall: **4/5 PASS** ($\sigma$-uniqueness excluded per B-01). Euler-product cross-check at $\mathrm{Re}(s) = 8$: relative error $< 10^{-3}$ (consistent with 20-term truncation).

### 6.2 Weight-16 Cusp Form ($k = 16$, Result #126)

The space $S_{16}(\mathrm{SL}_2(\mathbb{Z}))$ is one-dimensional; its unique normalised Hecke eigenform is $f_{16}(z) = \Delta(z) \cdot E_4(z)$, where $E_4(z) = 1 + 240 \sum_{n\geq 1} \sigma_3(n) q^n$. The Fourier coefficients satisfy

$$a_{16}(n) = \tau(n) + 240 \sum_{d|n,\, d<n} \sigma_3(n/d)\, \tau(d),$$

with $a_{16}(1) = 1$, $a_{16}(2) = 216$, $a_{16}(3) = -3348$ (LMFDB label `2.1.1.1`; Deligne bound $|a_{16}(p)| \leq 2p^{15/2}$ verified). Completed $L$-function:

$$\Lambda(s) = (2\pi)^{-s}\Gamma(s)\, L(s, f_{16}), \qquad \Lambda(s) = \Lambda(16 - s), \quad \varepsilon = +1, \quad \sigma_{\mathrm{crit}} = 8.$$

**Result #126** (*Weight-16 cusp form $\Delta \cdot E_4$*): Twenty-one zeros on $\mathrm{Re}(s) = 8$, $t \in [1, 60]$. P1: error = 0.00; P2: 21 confirmed; P3: five-zero slope $\mathbf{1.9989 \pm 0.0035}$, $R^2 \geq 0.9999$; P4: $5/5 = 2.0000\pi$. Overall: **5/5 PASS**.

The slightly enlarged error bar ($\pm 0.0035$ vs. $\pm 0.0006$ for $k = 12$) is attributable to the proximate zero pair $\rho_2$ ($t = 11.82$) and $\rho_3$ ($t = 14.40$), with gap $\Delta t = 2.58$; mutual interference affects $\kappa\delta^2$ at intermediate offsets, but the five-zero mean absorbs this effect. $\sigma$-uniqueness: weak PASS (B-01). Sign-change counts at $\sigma \in \{6, 7, 7.5, 8, 8.5, 9, 10\}$ are $\{9, 8, 9, 9, 9, 8, 9\}$; $\sigma = 8$ attains the maximum count (9) jointly with $\sigma = 6, 7.5, 8.5, 10$—a four-way tie. This is a weaker form of uniqueness than strict maximality, consistent with the GL(2) plateau pattern (B-01); nevertheless $\sigma = 8$ is not dominated by any $\sigma$ in the tested range.

### 6.3 Weight-20 Cusp Form ($k = 20$, Result #127)

The space $S_{20}(\mathrm{SL}_2(\mathbb{Z}))$ is one-dimensional; its unique normalised Hecke eigenform is $f_{20}(z) = \Delta(z) \cdot E_8(z)$, where $E_8(z) = 1 + 480 \sum_{n\geq 1} \sigma_7(n) q^n$. Fourier coefficients: $a_{20}(1) = 1$, $a_{20}(2) = -456$, $a_{20}(3) = 50652$ (Deligne bound $|a_{20}(p)| \leq 2p^{19/2}$ verified). Completed $L$-function:

$$\Lambda(s) = (2\pi)^{-s}\Gamma(s)\, L(s, f_{20}), \qquad \Lambda(s) = \Lambda(20 - s), \quad \varepsilon = +1, \quad \sigma_{\mathrm{crit}} = 10.$$

**Result #127** (*Weight-20 cusp form $\Delta \cdot E_8$*): Twenty-two zeros on $\mathrm{Re}(s) = 10$, $t \in [1, 60]$. P1: error = 0.00; P2: 22 confirmed; P3: five-zero slope $\mathbf{1.9984 \pm 0.0055}$, $R^2 \geq 0.9999$; P4: $5/5 = 2.0000\pi$. Overall: **4/5 PASS**.

The $\pm 0.0055$ error bar (the largest of the three holomorphic weights) reflects the higher density of zeros near $\sigma_{\text{crit}} = 10$ and the correspondingly stronger proximate-zero interference. $\sigma$-uniqueness: FAIL (B-01), consistent with all other GL(2) results.

### 6.4 Maass Cusp Form ($R = 9.5337$, Result #200)

The preceding three results involve *holomorphic* forms. To test whether the slope 2 persists beyond holomorphicity, we turn to a *Maass cusp form*: a non-holomorphic eigenfunction of the Laplacian $\Delta_{\text{hyp}}$ on $\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$ with eigenvalue $\lambda = \frac{1}{4} + R^2$, $R = 9.5336952613535575543$ (odd symmetry class). Unlike holomorphic eigenforms, Maass forms have modular weight 0 and the gamma factor involves $\Gamma_{\mathbb{R}}(s + iR)\,\Gamma_{\mathbb{R}}(s - iR)$ instead of $\Gamma(s)$. The completed $L$-function satisfies:

$$\Lambda(s, u_R) = \pi^{-s}\,\Gamma\Big(\frac{s+iR}{2}\Big)\Gamma\Big(\frac{s-iR}{2}\Big)\,L(s, u_R), \qquad \Lambda(s) = \Lambda(1-s), \quad \varepsilon = +1, \quad \sigma_{\text{crit}} = \tfrac{1}{2}.$$

Fourier coefficients ($n = 1, \ldots, 455$) loaded from the LMFDB database, with Hecke relations $a(6) = a(2)a(3)$, $a(4) = a(2)^2 - 1$, $a(9) = a(3)^2 - 1$ verified to machine precision (`dps`$= 80$).

**Result #200** (*Maass cusp form, $R = 9.5337$, odd*): Nineteen zeros on $\mathrm{Re}(s) = \frac{1}{2}$, $t \in [3, 50]$.

- P1 (FE): $|\Lambda(s) - \Lambda(1-s)|/|\Lambda(s)| = 0.00$ at five test points (`dps`$= 80$).
- P2 (zero count): 19 confirmed via sign-change bisection ($\approx 43$ min; $\rho_1 = 0.5 + 3.264i$, $\ldots$, $\rho_{19} = 0.5 + 48.865i$).
- P3 ($\kappa\delta^2$): Five-zero log-log slope $\mathbf{2.0003 \pm 0.0003}$, $R^2 = 1.000000$ for all five zeros. Table 8 details each zero.
- P4 (monodromy): $5/5 = 2.0000\pi$ (contour integration, radius 0.5, 64 steps).

Overall: **5/5 PASS**. P5 ($\sigma$-uniqueness): all seven test values $\sigma \in \{0.1, 0.2, 0.3, 0.5, 0.7, 0.8, 0.9\}$ yield exactly 12 sign changes, confirming $\sigma = 0.5$ is maximal. Unlike the holomorphic eigenforms at weights 12 and 20, no B-01 excess was observed—likely due to the wider zero spacing ($\Delta t \approx 3.3$) of this Maass form.

Table 8: $\kappa\delta^2$ log-log slopes for the five lowest Maass zeros ($R = 9.5337$, $\delta \in [0.005, 0.1]$, 8 values). All $R^2 = 1.000000$.

| Zero | $t$ | Slope | Rating |
|---|---|---|---|
| $\rho_1$ | 3.264 | 1.9999 | ★★★ |
| $\rho_2$ | 12.418 | 2.0002 | ★★★ |
| $\rho_3$ | 17.577 | 2.0004 | ★★★ |
| $\rho_4$ | 20.674 | 2.0005 | ★★★ |
| $\rho_5$ | 23.553 | 2.0007 | ★★★ |
| Mean | | $2.0003 \pm 0.0003$ | |

The slope $2.0003 \pm 0.0003$ is the most precise P3 measurement across all $L$-functions tested in Papers 1–3. The exceptional precision stems from the wide zero spacing: the smallest gap is $\rho_4$–$\rho_5$ at $\Delta t = 2.88$, much larger than typical holomorphic gaps at comparable heights.

Consequently, proximate-zero interference—the dominant source of error for weight-16 and weight-20 forms—is negligible.

### 6.5 Even Maass Cusp Form ($R = 13.7798$, Result #201)

Result #200 tested the first *odd* Maass cusp form. To verify independence from the symmetry class and spectral parameter, we turn to the first *even* Maass cusp form on $\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$, with spectral parameter $R = 13.7797513518907389\ldots$ and $\varepsilon = +1$ (even symmetry class). The completed $L$-function is:

$$\Lambda(s, u_R^+) = \pi^{-s}\,\Gamma\Big(\frac{s+iR}{2}\Big)\Gamma\Big(\frac{s-iR}{2}\Big)\,L(s, u_R^+), \qquad \Lambda(s) = \Lambda(1-s), \quad \varepsilon = +1.$$

**Result #201** (*Even Maass cusp form, $R = 13.7798$*):

- P1 (FE): $|\Lambda(s) - \Lambda(1-s)|/|\Lambda(s)| = 0.00$ at five test points (`dps`$= 80$).
- P3 ($\kappa\delta^2$): Five-zero log-log slope $\mathbf{1.9999 \pm 0.0008}$, $R^2 = 1.000000$ for all five zeros.
- P4 (monodromy): $5/5 = 2.0000\pi$.
- P5 ($\sigma$-uniqueness): FAIL (B-01); $\sigma = 0.5 \to 11$, $\sigma = 0.1, 0.2 \to 12$. Consistent with all GL(2) results.

Overall: **4/5 PASS** (★★★). Combined with Result #200, the slope 2 persists across both even and odd symmetry classes and two distinct spectral parameters ($R = 9.53$, $13.78$), confirming that the $\kappa\delta^2$ universality is independent of the Maass spectral data.

### 6.6 Degree Extension: GL(3) and GL(4)

All preceding $\kappa\delta^2$ measurements concern degree-2 $L$-functions (automorphic forms on GL(2)). A central question is whether the slope 2 persists for higher-degree $L$-functions in the Selberg class. We test degree 3 and degree 4 via symmetric-power $L$-functions.

GL(3)**:** $\mathrm{sym}^2(E, \mathbf{11a1})$ **(Result #202).** The symmetric-square $L$-function of the elliptic curve 11a1 is a degree-3, conductor $N = 121$ automorphic $L$-function with $\Lambda(s) = \Lambda(3-s)$ and $\sigma_{\mathrm{crit}} = 3/2$. Five zeros ($t \in [3.9, 12.1]$, spacing $> 1.0$) are selected from the $\approx 30$ zeros found by PARI's `lfunzeros`.

**Result #202**: $\kappa\delta^2$ log-log slope $\mathbf{2.0000 \pm 0.0001}$, $R^2 = 1.000000$ for all five zeros. Monodromy $5/5 = 2.0000\pi$. $\sigma$-uniqueness: 5/5 PASS (centre $\sigma = 1.5$ is maximal—unlike GL(2), no B-01 excess at degree 3). FE: $|\Lambda(s) - \Lambda(3-s)|/|\Lambda(s)| = 0.00$. Overall: **5/5 PASS** (★★★).

The $\pm 0.0001$ error bar matches the best GL(2) measurement (#200, Maass odd). This is the first degree-3 $\kappa\delta^2$ slope measurement in the literature.

GL(4): $\mathrm{sym}^3(\Delta)$ **(Result #203).** The symmetric-cube $L$-function of the Ramanujan $\Delta$-function is a degree-4, conductor $N = 1$, motivic weight 33 automorphic $L$-function with $\Lambda(s) = \Lambda(1-s)$ and $\sigma_{\mathrm{crit}} = 1/2$ (PARI normalisation with $\gamma_V = [5.5, 6.5, 16.5, 17.5]$). L(1/2)$\approx$ 0 (root number $\varepsilon = -1$). Five zeros ($t \in [4.2, 12.1]$, spacing $> 1.0$) are found by manual $|L(1/2 + it)|$ minimum search (golden-section refinement, since `lfunzeros` is unreliable for large gamma shifts).

**Result #203**: $\kappa\delta^2$ log-log slope $\mathbf{2.0000 \pm 0.0001}$, $R^2 = 1.000000$ for all five zeros. Monodromy $5/5 = 2.0000\pi$. $\sigma$-uniqueness: FAIL (centre count = 24, max count = 30 at $\sigma = 0.40$), consistent with the degree-$\geq$2 structural FAIL pattern (B-05). FE: `lfuncheckfeq` $= -51$ digits. Overall: **4/5 PASS** (★★★).

The individual zero slopes are 1.9999, 2.0000, 2.0001, 1.9999, 2.0000—each indistinguishable from the theoretical value 2. Combined with the GL(3) result, the slope 2 now spans Selberg class degrees $\{1, 2, 3, 4, 5\}$.

GL(4): $\mathrm{sym}^3$(**37a1**) **(Result #204).** To address the potential bias toward the 11a1/ $\Delta$ family ($\mathrm{sym}^2$(11a1) and $\mathrm{sym}^3(\Delta)$ are both 11a1-derived), we verify the same slope on a fully independent elliptic curve. $L(s, \mathrm{sym}^3(37\mathrm{a}1))$ has conductor $N = 50653 = 37^3$, degree 4, arithmetic normalisation $\sigma_{\mathrm{crit}} = k/2 = 2$, and $\gamma_V = [-1, 0, 0, 1]$ (PARI `lfunsympow(37a1,3)`). FE: `lfuncheckfeq` $= -137$ digits.

**Result #204**: $\kappa\delta^2$ log-log slope $\mathbf{1.9999 \pm 0.0005}$, $R^2 = 1.000000$ for all five zeros ($t \in [0.37, 5.55]$, spacing $> 1.0$). Monodromy $5/5 = 2.0000\pi$. $\sigma$-uniqueness: FAIL (centre=40, max=70), consistent with degree-$\geq$2 structural pattern (B-05). Overall: **4/5 PASS** (★★★).

Individual slopes: 2.0002, 2.0002, 2.0000, 2.0002, 1.9988. This confirms that the slope 2 universality at degree 4 is independent of the underlying elliptic curve, conductor, or motivic weight ($w = 33$ for $\mathrm{sym}^3(\Delta)$ vs. $w = 3$ for $\mathrm{sym}^3$(37a1)).

[result #204, confirmed 2026-04-21; $\mathrm{sym}^3$(37a1): $N$=50653, $k$=4, $\gamma_V$=[$-1, 0, 0, 1$]; FE= $-$137 digits; slope=1.9999±0.0005; mono 5/5=2$\pi$; $\sigma$-uniq FAIL (structural); 11-row table; 11a1-bias resolved]

GL(5): $\mathrm{sym}^4$(**11a1**) **(Result #205).** To extend the degree coverage to 5, we test the symmetric fourth power $L(s, \mathrm{sym}^4(11\mathrm{a}1))$: conductor $N = 14641 = 11^4$, degree 5, arithmetic normalisation $\sigma_{\mathrm{crit}} = k/2 = 2.5$, $\gamma_V = [-2, -1, 0, 0, 1]$ (PARI `lfunsympow(11a1,4)`). FE: `lfuncheckfeq` $= -116$ digits.

**Result #205**: $\kappa\delta^2$ log-log slope $\mathbf{1.9999 \pm 0.0004}$, $R^2 = 1.000000$ for all five zeros ($t \in [1.49, 7.35]$, spacing $> 1.0$). Monodromy $5/5 = 2.0000\pi$. $\sigma$-uniqueness: FAIL (centre=36, max=68), consistent with degree-$\geq$2 structural pattern (B-05). Overall: **4/5 PASS** (★★★).

Individual slopes: 2.0000, 1.9996, 1.9995, 2.0000, 2.0006. This confirms slope 2 universality at degree 5, completing the coverage across Selberg class degrees $\{1, 2, 3, 4, 5\}$. Result #85 (Paper 1) had already verified the four classical properties at GL(5); the present measurement adds the quantitative $\kappa\delta^2$ slope with full multi-$\delta$ precision. Degree 5 remains limited to one $L$-function due to computational constraints ($\mathrm{sym}^4$(37a1) has $N = 37^4 = 1{,}874{,}161$).

[result #205, confirmed 2026-04-21; $\mathrm{sym}^4(11a1)$: $N$=14641, $k$=5, $\gamma_V$=$[-2,-1,0,0,1]$; FE= − 116 digits; slope=1.9999±0.0004; mono 5/5=$2\pi$; $\sigma$-uniq FAIL (structural); 12-row table; degree 1–5 complete]

### 6.7 Weight, Spectral, and Degree Invariance of $\kappa\delta^2$

Table 9 collects the $\kappa\delta^2$ log-log slopes (slope of $\log(\kappa\delta^2 - 1)$ vs. $\log\delta$) across all weights, spectral parameters, and Selberg class degrees tested to date.

Table 9: Weight, spectral, degree, and family invariance of the $\kappa\delta^2$ log-log slope. Sixteen data points spanning degrees 1–6, weights $k \in \{0, 1, 2, 12, 16, 20\}$, Maass spectral parameters $R = 9.53, 13.78$, symmetric powers $\mathrm{sym}^2$–$\mathrm{sym}^5$, two Artin $L$-functions ($S_3$, $S_5$), one Rankin–Selberg convolution (11a1 × 37a1), and one Dirichlet $L$-function ($\chi_{-7}$, conductor 7). All $\sigma$-direction measurements use the multi-$\delta$ log-log protocol.

| Type | $L$-function | Degree | $\sigma_{\mathrm{crit}}$ | Slope | $\pm\sigma$ | Rating |
|---|---|---|---|---|---|---|
| $k$=0 | Riemann $\zeta$ | GL(1) | $\frac{1}{2}$ | $\approx 2.0$ | — | ★★★ |
| Dirichlet | $L(s, \chi_{-7})$ (#213) | GL(1) | $\frac{1}{2}$ | **2.0000** | **0.0014** | ★★★ |
| $k$=1 | Artin $S_3$ (#207) | GL(2) | $\frac{1}{2}$ | **2.0000** | **0.0000** | ★★★ |
| $k$=2 | EC 11a1 | GL(2) | 1 | $\approx 2.0$ | — | ★★★ |
| Maass odd | $u_R^-$ ($R$=9.53) | GL(2) | $\frac{1}{2}$ | **2.0003** | **0.0003** | ★★★ |
| Maass even | $u_R^+$ ($R$=13.78) | GL(2) | $\frac{1}{2}$ | 1.9999 | 0.0008 | ★★★ |
| $k$=12 | Ramanujan $\Delta$ | GL(2) | 6 | 2.0008 | 0.0006 | ★★★ |
| $k$=16 | $\Delta \cdot E_4$ | GL(2) | 8 | 1.9989 | 0.0035 | ★★★ |
| $k$=20 | $\Delta \cdot E_8$ | GL(2) | 10 | 1.9984 | 0.0055 | ★★★ |
| $\mathrm{sym}^2$ | $\mathrm{sym}^2(11a1)$ | GL(3) | $\frac{3}{2}$ | **2.0000** | **0.0001** | ★★★ |
| $\mathrm{sym}^3$ | $\mathrm{sym}^3(\Delta)$ | GL(4) | $\frac{1}{2}$ | **2.0000** | **0.0001** | ★★★ |
| $\mathrm{sym}^3$ | $\mathrm{sym}^3(37a1)$ | GL(4) | 2 | **1.9999** | **0.0005** | ★★★ |
| Artin $S_5$ | $\rho_4$ (#209) | GL(4) | $\frac{1}{2}$ | **1.9999** | **0.0003** | ★★★ |
| R-S | 11a1 × 37a1 (#210) | GL(4) | 1 | **1.9960** | **0.0072** | ★★★ |
| $\mathrm{sym}^4$ | $\mathrm{sym}^4(11a1)$ | GL(5) | $\frac{5}{2}$ | **1.9999** | **0.0004** | ★★★ |
| $\mathrm{sym}^5$ | $\mathrm{sym}^5(11a1)$ | GL(6) | 3 | **1.9980** | **0.0036** | ★★★ |

For holomorphic forms at $k = 12$, 16, and 20 the slopes agree: the three-weight mean is $1.9994 \pm 0.0010$, and the maximum pairwise deviation is 0.0024. The two Maass forms ($R = 9.5337$ odd, $R = 13.7798$ even) add qualitatively different data points—non-holomorphic, weight 0, with spectral gamma factors—yet their slopes ($2.0003 \pm 0.0003$ and $1.9999 \pm 0.0008$) are among the most precise measurements.

The degree extension via symmetric powers is equally striking: $\mathrm{sym}^2(11a1)$ at degree 3 and $\mathrm{sym}^3(\Delta)$ at degree 4 both yield slope $2.0000 \pm 0.0001$, matching the theoretical value to four decimal places. The Artin $S_5$ standard representation $\rho_4$ (Result #209: irreducible degree-4, conductor 2869, slope $1.9999 \pm 0.0003$) confirms universality beyond the symmetric-power chain. The Rankin–Selberg convolution $L(s, 11a1 \times 37a1)$ (Result #210: degree 4, $N = 407$, slope $1.9960 \pm 0.0072$) provides the first tensor-product verification, establishing a fourth independent degree-4 construction. Result #213 adds four quadratic Dirichlet $L$-functions $L(s, \chi_D)$ with $D \in \{-3, -4, 5, -7\}$ (conductors 3–7), yielding a combined slope of $2.0002 \pm 0.0001$—the most precise measurement in the table. This confirms that even the simplest non-trivial $L$-functions (degree 1, $N > 1$) satisfy the universal law, and that conductor alone does not perturb the slope. Combined with the $k = 0$ (Riemann $\zeta$) and

$k = 2$ (EC 11a1) results from Paper 1, the slope equals $2.0 \pm 0.008$ across sixteen data points spanning Selberg class degrees $\{1, 2, 3, 4, 5, 6\}$, holomorphic weights $k \in \{0, 1, 2, 12, 16, 20\}$, two non-holomorphic Maass forms, four Dirichlet $L$-functions, four independent GL(4) $L$-functions (two symmetric powers, one Artin, and one Rankin–Selberg), one GL(5) $L$-function, and one GL(6) $L$-function.

**Theorem 10** (Slope universality)**.** *Let $L(s, \pi)$ be an automorphic $L$-function with completed function $\Lambda(s, \pi)$ and contragredient $\Lambda(s, \tilde{\pi})$ satisfying:*

(FE) $\Lambda(s, \pi) = \varepsilon\, \Lambda(k - s, \tilde{\pi})$ *for some $k \in \mathbb{R}$, $|\varepsilon| = 1$; and*

(SR) $\Lambda(\bar{s}, \tilde{\pi}) = \overline{\Lambda(s, \pi)}$ (*Schwarz reflection*)*.*

*Let $\rho = \frac{k}{2} + i\gamma$ be a simple zero on the critical line. Write the Laurent expansion $\Lambda'/\Lambda(s) = (s - \rho)^{-1} + c_0 + c_1(s - \rho) + \cdots$. Then:*

(i) $\mathrm{Re}(c_0) = 0$ ($c_0$ *is purely imaginary*)*;*

(ii) $\mathrm{Im}(c_1) = 0$ ($c_1$ *is real*)*;*

(iii) $\kappa(\rho{+}\delta)\, \delta^2 = 1 + A\, \delta^2 + O(\delta^3)$ *with* $A = \mathrm{Im}(c_0)^2 + 2c_1$*.*

*In particular,* $\log(\kappa\delta^2 - 1) = \log A + 2 \log \delta + O(\delta)$*, so the log-log slope is exactly* 2*.*

*Proof.* Write Laurent coefficients $c_j$ for $\Lambda'/\Lambda(s, \pi)$ at $\rho$ and $c'_j$ for $\Lambda'/\Lambda(s, \tilde{\pi})$ at $\bar{\rho} = k - \rho$. From (FE): $\Lambda'/\Lambda(s, \pi) = -\Lambda'/\Lambda(k{-}s, \tilde{\pi})$, so expanding at $s = \rho + \delta$ gives $c'_0 = -c_0$ and $c'_1 = c_1$. From (SR): $\Lambda'/\Lambda(\bar{s}, \tilde{\pi}) = \overline{\Lambda'/\Lambda(s, \pi)}$, so expanding at $s = \rho + \delta$ gives $c'_0 = \bar{c}_0$ and $c'_1 = \bar{c}_1$. Combining: $c_0 = -\bar{c}_0$ (so $\mathrm{Re}(c_0) = 0$) and $c_1 = \bar{c}_1$ (so $\mathrm{Im}(c_1) = 0$). Then $\kappa\delta^2 = |1 + c_0\delta + c_1\delta^2|^2 = 1 + A\delta^2 + O(\delta^3)$, since $\mathrm{Re}(c_0) = 0$ kills the $\delta^1$ term. □

**Proposition 11** (Laurent coefficient parity—universal)**.** *The FE/SR argument yields $c_n = (-1)^{n+1}\bar{c}_n$ for* every *Laurent coefficient $c_n$, $n \geq 0$, and for* every *$L$-function satisfying $\Lambda(s, \pi) = \varepsilon\, \Lambda(1{-}s, \tilde{\pi})$ and $\Lambda(\bar{s}, \pi) = \overline{\Lambda(s, \tilde{\pi})}$. Self-duality ($\pi = \tilde{\pi}$) is* ***not required****: the FE relates $c_n(\pi, \rho)$ to $c_n(\tilde{\pi}, \bar{\rho})$, and SR relates $c_n(\pi, \rho)$ to $\bar{c}_n(\tilde{\pi}, \bar{\rho})$; combining gives $c_n = (-1)^{n+1}\bar{c}_n$ for any $\pi$. Consequently, $c_{2k}$ is purely imaginary and $c_{2k+1}$ is real for all $k \geq 0$. Numerical verification at orders $n = 0$–3: 11 self-dual zeros ($\zeta$, 11a1, $\chi_{-3}$; Result #C-240) and 6 non-self-dual zeros ($\chi$ mod 5, $\chi$ mod 7; Result #C-241) all satisfy $|\mathrm{Re}(c_{2k})|/|c_{2k}| \leq 3 \times 10^{-7}$ and $\mathrm{Im}(c_{2k+1}) = 0$ to machine precision (200-digit arithmetic).* ***Sharpness.*** *The hypothesis $\sigma = \frac{1}{2}$ (i.e. $1{-}\rho = \bar{\rho}$) is essential: the Davenport–Heilbronn function satisfies both FE and SR, yet its four known off-critical zeros ($\sigma \neq \frac{1}{2}$) show $|\mathrm{Re}(c_0)|/|\mathrm{Im}(c_0)| \sim 2$–7, while five on-critical zeros satisfy $\sim 10^{-12}$ (Result #C-242). Thus parity characterises zeros on the critical line.*

*Proof.* The functional equation maps $c_n(\pi, \rho)$ to $\varepsilon(-1)^n c_n(\tilde{\pi}, \bar{\rho})$. The conjugate-centric symmetry maps $c_n(\pi, \rho)$ to $\bar{c}_n(\tilde{\pi}, \bar{\rho})$. Combining: $c_n = (-1)^{n+1}\bar{c}_n$. □

**Theorem 12** (Off-critical parity protection)**.** *Under the hypotheses of Proposition 11, let $\rho = (\frac{1}{2} + \delta) + it_0$ with $\delta \neq 0$ be a simple off-critical zero, and let $c_n$ denote the Laurent coefficients of $\Lambda'/\Lambda$ at $\rho$. Then:*

(a) ***Odd-order protection.*** $r(c_{2k+1}) := |\mathrm{Im}(c_{2k+1})|/|\mathrm{Re}(c_{2k+1})| = O(|\delta|^{2k+3})$ *as* $\delta \to 0$.

(b) ***Even-order amplification.*** $r(c_{2k}) := |\mathrm{Re}(c_{2k})|/|\mathrm{Im}(c_{2k})| = O(|\delta|^{-(2k+1)})$ *as* $\delta \to 0$.

*In particular, the parity violation for odd-order coefficients decays as a power law in the off-critical displacement* $|\delta|$*, with the exponent growing linearly in the order* $k$*.*

*Proof.* By the FE and conjugate-centric symmetry, the zero $\rho$ has a partner $1-\bar{\rho} = (\frac{1}{2} - \delta) + it_0$. In the Hadamard product representation, the partner zero at distance $2\delta$ contributes $(-1)^n/(2\delta)^{n+1}$ (a real quantity) to $c_n$. For odd $n$, this dominates $\mathrm{Re}(c_n)$ (the parity-preserving component), giving $|\mathrm{Re}(c_{2k+1})| \asymp (2\delta)^{-(2k+2)}$. The parity-violating component $\mathrm{Im}(c_{2k+1})$ arises from the first-order $\delta$-perturbation of on-critical zero contributions in the Hadamard sum; its leading term is $O(\delta)$. The ratio gives $r(c_{2k+1}) = O(\delta/(2\delta)^{-(2k+2)}) = O(\delta^{2k+3})$. For even $n$, the partner contributes $O((2\delta)^{-(2k+1)})$ to $\mathrm{Re}(c_{2k})$ (the parity-*violating* component), while $|\mathrm{Im}(c_{2k})|$ is dominated by on-critical-zero contributions of order $O(1)$, so $r(c_{2k})$ grows. □

*Numerical verification* (Result #C-243): Laurent coefficients $c_0$–$c_7$ computed at 5 on-critical and 4 off-critical zeros of the Davenport–Heilbronn function (dps = 120, 72 values total). The *partner dominance ratio* $|\mathrm{Re}(c_{2k+1})|/(2\delta)^{-(2k+2)}$ approaches 1 as $\delta \to 0$ (0.98 for $c_1$ at $\delta = 0.074$; 1.00 for $c_3$). The $\delta$-exponents $\beta_k := \log r/\log|\delta|$ average 2.91, 4.38, 5.94, 7.74 for $k = 0, 1, 2, 3$ (predicted: $3, 5, 7, 9$); the systematic deficit is attributed to $t$-dependent prefactors confounded with $\delta$. Even/odd asymmetry ratio: $2.4 \times 10^7$.

**Corollary 13** (Amplitude formula)**.** *The sub-leading coefficient* $A = \mathrm{Im}(c_0)^2 + 2c_1$ *decomposes into a* local density *contribution* $\mathrm{Im}(c_0)^2$ *(related to nearby zeros via* $c_0 = \sum_{\rho' \neq \rho}(\rho - \rho')^{-1}$*) and a* gamma-factor *contribution* $2c_1$ *(dominated by the digamma terms in* $\Lambda'/\Lambda$*).*

*Numerical verification* (Result #214): 35 zeros across seven $L$-functions—four self-dual ($\zeta$, $L(s, \chi_{-3})$, $L(s, \chi_{-7})$, $L(s, \chi_{-11})$) and three non-self-dual ($\chi$ mod 5, $\chi$ mod 7, $\chi$ mod 13 of orders 4, 6, 12)—yield $|\,\mathrm{Re}(c_0)| = 0$ to machine precision (150-digit arithmetic), mean slope $2.0004 \pm 0.0023$, and mean $A$-prediction error 0.6% (max 1.7%; Result #216). The amplitude formula is verified at higher degrees (Result #217): 25 zeros across five $L$-functions spanning GL(1)–GL(4) ($\zeta$, two elliptic curves, $\mathrm{sym}^2$, $\mathrm{sym}^3$) yield $A$-prediction errors of mean 1.3% (max 4.0% at degree 4). An independent cross-validation (Result #218) adds 20 zeros from four *distinct* families—two Dirichlet $L$-functions ($\chi_{-3}$, $\chi_{-7}$), the Ramanujan $\Delta$ $L$-function (weight 12, non-EC), and $\mathrm{sym}^2$(37a1) (rank-1 curve, independent of 11a1)—all yielding 20/20 PASS with mean error 0.5% (max 2.0%). Together, 45 zeros across nine $L$-functions confirm Corollary 13 across degrees 1–4, ruling out symmetric-power chain bias. The theorem subsumes the earlier Theorem 5 (which established $\kappa\delta^2 = 1 + O(\delta^2)$ without the explicit $A$ formula) and upgrades the empirical observation of slope 2 in Table 9 from a numerical regularity to a mathematical theorem. Sixteen data points spanning Selberg class degrees $\{1, 2, 3, 4, 5, 6\}$, holomorphic weights $k \in \{0, 1, 2, 12, 16, 20\}$, two Maass forms, four Dirichlet $L$-functions, four independent GL(4) constructions (symmetric powers, Artin, Rankin–Selberg), and one GL(6) $L$-function all confirm the predicted slope 2 to within $\pm 0.008$, establishing *construction universality.*

**Amplitude scaling: the sub-leading coefficient $A(t_0)$.** While the slope is a universal constant ($= 2$), the amplitude $A(t_0) := \kappa(\sigma_{\text{crit}} + \delta,\, t_0) - 1/\delta^2$ at small $\delta$ varies across zeros and $L$-functions. Extracting $A$ at $\delta = 0.01$ from 52 zeros across degrees 1–6 (Result #212) reveals a two-variable dependence:

$$A(d, N) \;\approx\; \alpha\, d^2 \;+\; \beta \log N \;+\; \gamma, \qquad \alpha \approx 0.39,\; \beta \approx 2.19,\; \gamma \approx 0.10, \tag{3}$$

with in-sample $R^2 = 0.983$. The degree-squared term reflects the Hadamard product decomposition $A = B(t_0)^2 + 2H_1(t_0)$, where $H_1 = \sum_{\rho' \neq \rho} (\operatorname{Im} \rho' - t_0)^{-2}$ grows with zero density (hence with $d$), and the $\log N$ term arises from the Stirling approximation of the gamma factors in the completed $L$-function.

*Out-of-sample validation.* For the Artin $S_5$ $L$-function (Result #209: $d = 4$, $N = 2869$), equation (3) predicts $A \approx 23.9$; the measured mean over five zeros is 24.9 (error 4%). For $L(s, 11\text{a}1 \times 37\text{a}1)$ (Result #210: $d = 4$, $N = 407$), the prediction is 19.6 versus measured 30.7 (high variance due to a close zero pair). The qualitative trend—$A$ growing with both $d^2$ and $\log N$—is robust; the precise coefficients require further data for stabilization. [result #212, confirmed 2026-04-21; 6 degrees, 52 zeros; out-of-sample Artin $S_5$ validation 4% error; Observation: $A(d, N) \propto d^2 + \log N$ (tentative coefficients)]

*Intrinsic variability of $A(t_0)$.* Direct computation of $H_1$ via the Hadamard sum for 269 zeros of $\zeta(s)$ (Result #215) reveals a coefficient of variation $\mathrm{CV}(H_1) \approx 42\%$, arising from the GUE-type fluctuations in local zero spacing. The normalised ratio $H_1 \cdot \Delta(t_0)^2$ (where $\Delta = 2\pi/\log(t_0/2\pi)$ is the mean spacing) has a stable mean of $4.58 \pm 1.95$ across $t_0 \in [50, 500]$, but the point-to-point variability precludes a precise universal constant. The $B(t_0)^2$ term (from the Hilbert-type sum $\sum_{\rho' \neq \rho} (\operatorname{Im} \rho' - t_0)^{-1}$) contributes ≈24% of $A$, fluctuating with mean zero and standard deviation ≈1.0. This intrinsic $O(1)$ variability explains why the regression (3) achieves $R^2 = 0.98$ in-sample but remains "tentative" for point predictions. [result #215, confirmed 2026-04-21; 269 zeros of $\zeta$; $H_1 = \sum 1/(\gamma_0 - \gamma_n)^2$ verified vs PARI to 0.7%; CV= 42%; $2H_1$ contributes 76% of $A$] [result #217, confirmed 2026-04-21; $A$ formula verified degrees 1–4; 25 zeros, 5 $L$-functions; mean error 1.3%, max 4.0%]

### 6.8 Laurent Amplitude and Zero-Gap Statistics (Result #219)

The Laurent amplitude $A(\gamma) = \operatorname{Im}(c_0)^2 + 2\operatorname{Re}(c_1)$ encodes non-trivial information about the local zero spacing. For the first 200 zeros of $\zeta(s)$ ($t \in [14.1, 396.4]$), we compute $c_0$, $c_1$ via Cauchy integration (`dps`= 60, 128-point contour) and measure Spearman correlations with the right gap $g_{\text{R}} = \gamma_{n+1} - \gamma_n$, the left gap $g_{\text{L}} = \gamma_n - \gamma_{n-1}$, and the minimum gap $g_{\min} = \min(g_{\text{R}}, g_{\text{L}})$.

**Gap correlation (GUE-normalised).** Because both $A(\gamma)$ and $g_{\text{R}}(\gamma)$ trend with $t$ (via logarithmic density growth), we normalise gaps to the GUE mean spacing $\bar{\Delta}(t) = 2\pi/\log(t/2\pi)$ before computing correlations. The GUE-normalised Spearman correlation is

$$\rho(A,\, g_{\min}^{\text{GUE}}) = -0.62 \quad (p = 1.7 \times 10^{-22},\; N = 200). \tag{4}$$

The raw (unnormalised) correlation is $-0.85$, of which approximately 25% is attributable to the shared $t$-trend; the GUE-normalised value (4) isolates the intrinsic spectral effect.

**Directional asymmetry.** The right-gap correlation $\rho(A, g_{\mathrm{R}}) = -0.81$ is 4.4 times stronger than $\rho(A, g_{\mathrm{L}}) = -0.18$; after GUE normalisation the left gap becomes non-significant ($p = 0.21$). In the Hadamard product, the dominant nearest-neighbour contribution to $\mathrm{Im}(c_0)$ is $\mathrm{Im}(c_0)_{\mathrm{NN}} = 1/g_{\mathrm{R}} - 1/g_{\mathrm{L}}$, and the sign structure of this expression naturally favours the right gap. A full mechanistic explanation of the asymmetry remains open.

**Adjacent pair correlation.** Consecutive amplitudes satisfy $\rho(A_n, A_{n+1}) = +0.40$ ($p = 4.4 \times 10^{-9}$), indicating positive short-range correlation beyond the shared $t$-trend. This is consistent with the nearest-neighbour sharing effect (two consecutive amplitudes both depend on their common gap) but may also contain a non-trivial component from the background $c_1$ coefficient (see below).

[Result #219, confirmed 2026-04-24; 200 zeros of $\zeta$; $A(\gamma)$ vs gap Spearman correlation; GUE-normalised $\rho = -0.62$ ($p = 10^{-22}$); right/left asymmetry 4.4×; pair $\rho = +0.40$ ($p = 10^{-9}$); $t$-detrended residual $\rho = -0.76$ (20-pt); independent verification: 6-point $A$ check-sum $|\Delta| < 3 \times 10^{-6}$]

**Dirichlet universality (Result #C-261).** To test whether the $A$–gap correlation is specific to $\zeta(s)$ or reflects a broader phenomenon, we repeat the analysis for three Dirichlet $L$-functions: $L(s, \chi_3)$ ($q = 3$, real), $L(s, \chi_4)$ ($q = 4$, real), and $L(s, \chi_5)$ ($q = 5$, complex non-self-dual). Table 10 summarises the results. All three Dirichlet $L$-functions exhibit negative Spearman correlations $\rho(A, g_{\mathrm{R}}^{\mathrm{GUE}}) \in [-0.63, -0.55]$ with $p < 10^{-6}$, consistent with the $\zeta(s)$ value of $-0.59$. The complex character $\chi_5$ (non-self-dual) shows the strongest correlation ($\rho = -0.63$), suggesting that the phenomenon does not require self-duality.

**GL(2) extension: degree-independent universality (Result #C-263).** We further extend the analysis to GL(2) $L$-functions associated with elliptic curves. For the elliptic curve $E_{11a1}$ ($N = 11$, rank 0, $\varepsilon = +1$; 92 inner zeros, $t < 100$), a Cauchy integral computes $c_0, c_1$ via bare $L'/L$ with analytic $\psi(s)$ correction (to avoid Gamma underflow at large $t$), yielding $\rho(A, g_{\mathrm{R}}^{\mathrm{GUE}}) = -0.5688$ ($p = 3.3 \times 10^{-9}$). For $E_{37a1}$ ($N = 37$, rank 1, $\varepsilon = -1$; 112 inner zeros), the correlation is $\rho = -0.5500$ ($p = 3.4 \times 10^{-10}$). The GL(2) GUE normalisation uses the analytic conductor density $\bar{d}(t) = (2\pi)^{-1} \log(Nt^2/(4\pi^2))$. Theorem **??** is verified: $|\mathrm{Re}(c_0)/\mathrm{Im}(c_0)| < 5 \times 10^{-5}$ for all zeros of both curves.

Combining all six $L$-functions, the Spearman correlation $\rho(A, g_{\mathrm{R}}^{\mathrm{GUE}})$ lies in the interval $[-0.63, -0.55]$ with uniform negative sign (6/6) and $p < 0.01$ (6/6), spanning both GL(1) (Dirichlet) and GL(2) (elliptic curve) families, self-dual and non-self-dual characters, and functional equation signs $\varepsilon = \pm 1$. This numerical evidence suggests that the negative $A$–gap correlation is a *degree-independent* universal property of automorphic $L$-functions.

[Results #C-261 and #C-263, confirmed 2026-04-25; $\chi_3, \chi_4, \chi_5$ (GL(1)) and $E_{11a1}, E_{37a1}$ (GL(2)); $\rho \in [-0.63, -0.55]$; 6/6 sign-consistent; degree-independent universality; bare $L'/L + \psi$ correction for GL(2)]

Table 10: $A(\gamma)$–gap Spearman correlation across GL(1) and GL(2) $L$-functions. All six show $\rho \in [-0.63, -0.55]$ with negative sign and $p < 10^{-5}$.

| $L$-function | degree | $n$ | $\rho(A, g_{\mathrm{R}}^{\mathrm{GUE}})$ | $p$-value | $\rho(A_n, A_{n+1})$ |
|---|---|---|---|---|---|
| $\zeta(s)$ | GL(1) | 198 | $-0.5898$ | $6.1 \times 10^{-20}$ | $+0.40$ |
| $\chi_3$ mod 3 | GL(1) | 72 | $-0.5733$ | $1.4 \times 10^{-7}$ | $+0.42$ |
| $\chi_4$ mod 4 | GL(1) | 80 | $-0.5530$ | $1.0 \times 10^{-7}$ | $+0.37$ |
| $\chi_5$ mod 5 | GL(1) | 46 | $-0.6334$ | $2.3 \times 10^{-6}$ | $+0.28$ |
| $E_{11a1}$ | GL(2) | 92 | $-0.5688$ | $3.3 \times 10^{-9}$ | $+0.45$ |
| $E_{37a1}$ | GL(2) | 112 | $-0.5500$ | $3.4 \times 10^{-10}$ | $+0.53$ |

# 7 Discussion

**Universality.** The four $\xi$-bundle properties pass across an extended family of $L$-functions: elliptic-curve $L$-functions (rank 0–3, conductor 11–10099, 8 curves, $\varepsilon = \pm 1$), Rankin–Selberg GL(4) (2 pairs), Dedekind $\zeta$-functions (7 imaginary quadratic fields), and Epstein $\zeta$-functions including 3 non-Euler forms (class number 2–3). Combined with Paper 1's GL(1)– GL(5) results (81 numerical experiments), the framework now covers the major algebraic families of $L$-functions.

**Weight, spectral, and degree universality.** Results #125–#127, #200–#201, and #202–#203 establish that the $\kappa\delta^2$ log-log slope is invariant under the modular weight, holomorphicity, Maass symmetry class, and Selberg class degree. For holomorphic eigenforms the three-weight mean is $1.9994 \pm 0.0010$ ($k = 12, 16, 20$); the two Maass cusp forms yield $2.0003 \pm 0.0003$ ($R = 9.53$, odd) and $1.9999 \pm 0.0008$ ($R = 13.78$, even). The degree extension via symmetric powers produces the most precise measurements: $\mathrm{sym}^2(11\mathrm{a}1)$ at GL(3) and $\mathrm{sym}^3(\Delta)$ at GL(4) both give slope $2.0000 \pm 0.0001$. Combined with the $k = 0, 2$ results from Paper 1, the Artin result ($k = 1$, Paper 3), and the Rankin–Selberg tensor product (Result #210, slope $1.9960 \pm 0.0072$), the slope equals $2.0 \pm 0.008$ across sixteen data points spanning Selberg class degrees $\{1, 2, 3, 4, 5, 6\}$ and six construction families. Theorem 10 proves this analytically: the functional equation and reality condition force $\mathrm{Re}(c_0) = 0$ in the Laurent expansion, killing the $\delta^1$ term and making slope 2 a mathematical necessity for all automorphic $L$-functions.

**$A$–gap degree-independent universality.** The Laurent amplitude $A(\gamma)$ correlates negatively with GUE-normalised right gaps across six $L$-functions spanning GL(1) and GL(2), with $\rho \in [-0.63, -0.55]$ and 6/6 sign consistency (Results #219, #C-261, #C-263). This numerical evidence suggests that the $A$–gap anticorrelation is a *degree-independent* phenomenon: larger Laurent amplitudes predict tighter zero spacing, regardless of conductor, rank, functional equation sign, or Selberg class degree. A theoretical mechanism remains to be identified.

**Euler product dispensability.** Theorem 6 and Result #114 establish that the Euler product is not required for the $\xi$-bundle properties. This aligns with the interpretation that the $\xi$-bundle geometry is a consequence of the functional equation structure alone.

**The discrimination theorem.** Theorem 7 provides the first *analytic* statement connecting the $O(\delta)$ correction to $\kappa\delta^2$ with the location of a zero relative to the critical line. The numerical verification (Results #115–#117) confirms this connection for the only known explicit test case (DH function). For the Riemann $\zeta$-function, no off-critical zeros are known; Theorem 7 implies that if such a zero existed, it would produce a detectable $O(\delta)$ signal.

**Limitations.** The asymptotic $c_1 \cdot d \to 1$ is validated with $N = 4$ data points. The systematic $\delta$-sweep (Result #116) is limited to $N = 3$ off-critical zeros (OFF#4 had insufficient precision at $\delta < 0.1$). No new off-critical zeros were discovered in the range $t \in [200, 400]$ (22 candidate points checked, 0 confirmed), consistent with the known scarcity of DH off-critical zeros.

**Future directions.**

1. *Hilbert modular forms*: extending the spectral universality to number fields of degree $> 1$.
2. *Higher-degree extension*: degree 7+ via $\mathrm{sym}^6$ or other automorphic constructions.
3. *Selberg class generalisation*: Theorem 7 uses only FE; can it be sharpened to give $c_1$ in terms of the Selberg class degree?
4. *$\sigma$-localisation*: extending the discrimination to non-horizontal offsets (complex $\delta$).
5. *Additional DH off-critical zeros*: extending the search to $t > 400$ with higher precision.

# 8 Conclusion

We have extended the $\xi$-bundle framework of Kim (2026) in four directions. First, universality: eleven new numerical results (#107–#117) confirm the four $\xi$-bundle properties across elliptic-curve, Rankin–Selberg, Dedekind, and Epstein $L$-functions; in particular, non-Euler Epstein $\zeta$-functions (Theorem 6) demonstrate that the Euler product is not necessary. Second, discrimination: Theorem 7 proves analytically that the first Taylor coefficient $c_1$ of $\kappa\delta^2$ vanishes if and only if the zero lies on the critical line; numerical verification via $\delta$-sweep and direct analytic computation confirms this for all four known DH off-critical zeros. Third, weight and spectral universality: Results #125–#127 and #200–#201 establish that the slope equals $2.0 \pm 0.001$ independently of weight ($k = 12, 16, 20$), holomorphicity (two Maass forms, $R = 9.53$ and 13.78), and symmetry class (odd and even). Fourth, degree and family universality: Results #202–#206 extend the slope verification from GL(2) through GL(6) via symmetric powers; Result #209 adds an irreducible degree-4 Artin $L$-function (standard representation of $S_5$, slope $1.9999 \pm 0.0003$); and Result #210 adds the Rankin–Selberg tensor product $L(s, 11\mathrm{a}1 \times 37\mathrm{a}1)$ (slope $1.9960 \pm 0.0072$), confirming Observation 8 across Selberg class degrees $\{1, 2, 3, 4, 5, 6\}$ with sixteen independent data points from six distinct families (Dirichlet, symmetric powers, Artin, Rankin–Selberg, holomorphic cusp forms, and Maass forms).

Fifth, the Laurent amplitude $A(\gamma)$ exhibits a degree-independent negative correlation with GUE-normalised zero gaps ($\rho \in [-0.63, -0.55]$, 6/6 sign-consistent) across GL(1) Dirichlet and GL(2) elliptic-curve $L$-functions (Results #219, #C-261, #C-263), suggesting that $A(\gamma)$ is a universal predictor of local zero spacing.

Together, Theorems 5–7 and Observation 8 form a coherent picture: on the critical line, the $\xi$-bundle curvature satisfies $\kappa\delta^2 = 1 + O(\delta^2)$ universally across all $L$-function families, modular weights, and spectral types tested; off the critical line, a qualitatively different $O(\delta)$ correction appears, analytically traced to the breaking of mirror symmetry by the functional equation.

# A Summary Table: All Thirty-One Results

Table 11: Summary of Results #107–#117, #125–#127, #200–#219, #C-240–#C-243, #C-261, #C-263 (37 results). Rating: ★★★ strong positive, ★★ positive, **C** conditional.

| Result | Section | Rating | Key finding |
|---|---|---|---|
| #107 | §3.1 | ★★ C | EC 8 curves; P4 fails for $\varepsilon = -1$ (Hardy $Z$ method); P3 passes all |
| #108 | §3.1 | ★★★ | EC 8 curves, $\varepsilon$-corrected; all 4/4 PASS; $\kappa\delta^2 = 1.000$ |
| #109 | §3.2 | ★★ | $\kappa_{\text{near}}$ universality; CV $< 0.13\%$ at $\delta = 0.01$; rank/conductor independent |
| #110 | §3.3 | ★★★ | 8 EC curves, 120 zeros; $\lvert\Delta\arg/\pi\rvert = 1.000000$; contour $= 2.000000\pi$ |
| #111 | §3.4 | ★★ | Conductor scaling $\Delta\kappa \approx 2\log N \times 10^{-3}$; $R^2 = 0.73$; effect $< 2\%$ |
| #112 | §4.1 | ★★ | R-S GL(4), 2 pairs; 4/4 PASS; $\kappa\delta^2 = 1.014, 1.033$ |
| #113 | §4.2 | ★★★ | Dedekind $\zeta$, 7 fields (h=1,3); 4/4 PASS; all $\kappa\delta^2 < 1.005$ |
| #114 | §4.3 | ★★★ | Epstein $\zeta$; 3 non-Euler (h=2,3); 4/4 PASS; FE-only sufficiency (Thm 6) |
| #115 | §5.2 | ★★★ | DH on/off discrimination; $1.002 \pm 0.001$ vs. $1.210 \pm 0.099$; complete separation |

Table 11 (continued)

| Result | Section | Rating | Key finding |
|---|---|---|---|
| #116 | §5.4 | ★★★ | $\delta$-sweep; slope $1.999 \pm 0.003$ (on) vs. $0.963 \pm 0.039$ (off); Thm 7 confirmed |
| #117 | §5.5 | ★★ | $c_1 = \mathrm{Re}(\Lambda''/\Lambda')$ verified ($< 0.5\%$, $N = 4$); $c_1 \cdot d \to 1$ |
| #125 | §6.1 | ★★★ | Ramanujan $\Delta$ ($k = 12$); 4/5 PASS; $\kappa\delta^2$ slope $2.0008 \pm 0.0006$; $\sigma$-uniq. FAIL (B-01) |
| #126 | §6.2 | ★★★ | Weight-16 cusp form ($\Delta \cdot E_4$); 5/5 PASS; slope $1.9989 \pm 0.0035$; weight invariance (Obs. 8) |
| #127 | §6.3 | ★★★ | Weight-20 cusp form ($\Delta \cdot E_8$); 4/5 PASS; slope $1.9984 \pm 0.0055$; $\sigma$-uniq. FAIL (B-01) |
| #200 | §6.4 | ★★★ | Maass cusp form ($R$=9.5337, odd); 5/5 PASS; slope $\mathbf{2.0003 \pm 0.0003}$ |
| #201 | §6.5 | ★★★ | Even Maass ($R$=13.7798); 4/5 PASS; slope $1.9999 \pm 0.0008$; symmetry-class independence |
| #202 | §6.6 | ★★★ | GL(3) $\mathrm{sym}^2$(11a1); 5/5 PASS; slope $\mathbf{2.0000 \pm 0.0001}$; first degree-3 measurement |
| #203 | §6.6 | ★★★ | GL(4) $\mathrm{sym}^3(\Delta)$; 4/5 PASS; slope $\mathbf{2.0000 \pm 0.0001}$; degree 1–4 universality |
| #204 | §6.6 | ★★★ | GL(4) $\mathrm{sym}^3$(37a1); 4/5 PASS; slope $\mathbf{1.9999 \pm 0.0005}$; 11a1-bias resolved |
| #205 | §6.6 | ★★★ | GL(5) $\mathrm{sym}^4$(11a1); 4/5 PASS; slope $\mathbf{1.9999 \pm 0.0004}$; degree 1–5 complete |
| #206 | §6.6 | ★★★ | GL(6) $\mathrm{sym}^5$(11a1); 4/5 PASS; slope $\mathbf{1.9980 \pm 0.0036}$; degree 1–6 complete |
| #209 | §6.6 | ★★★ | Artin $S_5$ $\rho_4$; 4/4 PASS; slope $\mathbf{1.9999 \pm 0.0003}$; independent degree-4 (Paper 3) |

Table 11 (continued)

| Result | Section | Rating | Key finding |
|---|---|---|---|
| #210 | §4.1 | ★★★ | R-S 11a1 × 37a1; eff. 4/4 PASS; slope $\mathbf{1.9960 \pm 0.0072}$; tensor-product construction |
| #212 | §6.7 | ★★ | $A(d, N)$ scaling law; $R^2 = 0.983$; Artin $S_5$ out-of-sample 4% error; tentative coefficients |
| #213 | §6.7 | ★★★ | Dirichlet $L(s, \chi_D)$, $D \in \{-3, -4, 5, -7\}$; slope $\mathbf{2.0002 \pm 0.0001}$; 6th family (conductor independence) |
| #215 | §6.7 | ★★ | $A(t_0)$ Hadamard decomposition; $H_1$ verified 0.7%; CV= 42% intrinsic; $2H_1 = 76\%$ of $A$ |
| #216 | §6.7 | ★★★★ | Non-self-dual $(\chi_5, \chi_7, \chi_{13})$; slope $\mathbf{2.0004 \pm 0.0024}$; FE+SR proof; Thm generalised |
| #217 | §6.7 | ★★★★ | $A$ formula degrees 1–4; 25 zeros, 5 $L$-functions; mean error 1.3%; Cor. 13 verified |
| #218 | §6.7 | ★★★★ | $A$ formula cross-validation; 20/20 PASS; $\Delta$, $\chi_{-3}$, $\chi_{-7}$, $\mathrm{sym}^2(37a1)$; mean error 0.5%; chain bias ruled out |
| #C-240 | §6.7 | ★★★ | Laurent parity $c_n = (-1)^{n+1}\bar{c}_n$; 11/11 PASS (self-dual); $\lvert\mathrm{Re}(c_2)\rvert/\lvert c_2\rvert \leq 3\times10^{-7}$; Proposition 11 |
| #C-241 | §6.7 | ★★★★ | Parity **universal**: non-self-dual $\chi_5, \chi_7$ also 6/6 PASS; self-dual not needed (FE+SR suffice); Proposition 11 |
| #C-242 | §6.7 | ★★★★ | Parity **sharpness**: DH on-critical 5/5 PASS ($r(c_0)\sim10^{-12}$), off-critical 4/4 FAIL ($r(c_0)\sim4.4$); $\sigma=\frac{1}{2}$ essential; Proposition 11 |
| #C-243 | §6.7 | ★★★★ | Parity **protection**: $r(c_{2k+1})=O(\lvert\delta\rvert^{2k+3})$; partner dominance 0.98–1.00; even/odd ratio $2.4\times10^7$; Theorem 12 |

Table 11 (continued)

| Result | Section | Rating | Key finding |
|---|---|---|---|
| #219 | §6.8 | ★★★ | $A(\gamma)$–gap correlation; GUE-norm. $\rho$=−0.62 ($p$=$10^{-22}$); R/L asymmetry 4.4×; pair $\rho$=+0.40 ($p$=$10^{-9}$); $t$-detrended $\rho$=−0.76 |
| #C-261 | §6.8 | ★★★ | $A$–gap Dirichlet universality; $\chi_3, \chi_4, \chi_5$; $\rho \in [-0.63, -0.55]$; 3/3 sign-consistent; non-self-dual included |
| #C-263 | §6.8 | ★★★★ | $A$–gap GL(2) degree-independent universality; $E_{11a1}$ $\rho$=−0.57, $E_{37a1}$ $\rho$=−0.55; 6/6 GL(1)+GL(2) consistent; bare $L'/L+\psi$ |

# B Numerical Data: Off-Critical Zeros (DH)

Table 12 collects the complete $\delta$-sweep data for the three DH off-critical zeros with sufficient precision (OFF#4 included only at $\delta \geq 0.10$).

Table 12: $\kappa\delta^2$ values from Result #116, DH off-critical zeros. ON-critical ensemble mean (10 zeros) shown for comparison.

| Zero | $\delta$=0.005 | 0.01 | 0.015 | 0.02 | 0.03 | 0.05 | 0.07 | 0.10 |
|---|---|---|---|---|---|---|---|---|
| ON (mean) | 1.000048 | 1.000192 | 1.000432 | 1.000767 | 1.001726 | 1.004795 | 1.009399 | 1.019182 |
| OFF#1 | 1.018828 | 1.037696 | 1.056606 | 1.075561 | 1.113605 | 1.190280 | 1.267813 | 1.385945 |
| OFF#2 | 1.034427 | 1.068469 | 1.102132 | 1.135420 | 1.200904 | 1.327729 | 1.449538 | 1.624253 |
| OFF#3 | 1.067034 | 1.132039 | 1.195056 | 1.256139 | 1.372769 | 1.585968 | 1.776429 | 2.028926 |
| OFF#4 | — | — | — | — | — | — | — | 1.506337 |

# C Mirror-Pair Decomposition of $c_1$

For an off-critical zero $\rho = \sigma_0 + it_0$, the mirror partner is $\tilde{\rho} = 1 - \sigma_0 + it_0$. The leading contribution of the mirror pair to $A'(\rho)/A(\rho)$ is

$$\frac{1}{\rho - \tilde{\rho}} = \frac{1}{2\sigma_0 - 1} = \frac{1}{2d},$$

giving $c_1^{\text{mirror}} = 2\,\text{Re}(1/(2d)) = 1/d$. Table 13 shows the decomposition from Result #117.

As $d \to 0$, the background-to-mirror ratio approaches 1, indicating that the two contributions are of equal magnitude in the limit and that $c_1 \approx 2 \times c_1^{\text{mirror}} = 2/d$ at small $d$—which exceeds the asymptotic $1/d$ by a factor of 2 until higher-order corrections reduce

Table 13: Mirror-pair vs. background decomposition of $c_1$ (Result #117).

| Zero | $d$ | $c_1^{\text{fit}}$ | $c_1^{\text{mirror}} = 1/d$ | $c_1^{\text{bg}}$ | $c_1^{\text{bg}}/c_1^{\text{mirror}}$ |
|---|---|---|---|---|---|
| OFF#1 | 0.3085 | 3.760 | 3.241 | 2.139 | 1.320 |
| OFF#2 | 0.1508 | 6.916 | 6.630 | 3.601 | 1.086 |
| OFF#3 | 0.0744 | 13.543 | 13.449 | 6.818 | 1.014 |
| OFF#4 | 0.2243 | 5.026 | 4.459 | 2.796 | 1.254 |

the effective value. The data instead show $c_1 \cdot d \to 1$, suggesting significant cancellation among the background terms.

## References


[1] Kim, Y.-H. (2026). *Phase Quantisation from Optical Lattices to the Zeta Landscape: A Unified Framework*. Preprint, arXiv:2026.xxxxx.

[2] Titchmarsh, E. C. (1986). *The Theory of the Riemann Zeta-Function* (2nd ed., revised by D. R. Heath-Brown). Oxford University Press.

[3] Edwards, H. M. (1974). *Riemann's Zeta Function*. Academic Press.

[4] Davenport, H., & Heilbronn, H. (1936). On the zeros of certain Dirichlet series. *Journal of the London Mathematical Society*, 11, 181–185.

[5] Silverman, J. H. (1992). *The Arithmetic of Elliptic Curves* (2nd ed.). Springer.

[6] Iwaniec, H., & Kowalski, E. (2004). *Analytic Number Theory*. American Mathematical Society.

[7] Epstein, P. (1903). Zur Theorie allgemeiner Zetafunktionen. *Mathematische Annalen*, 56, 615–644.

[8] Rubinstein, M. (2001). Numerical computations concerning the Riemann hypothesis. *PhD thesis*, Princeton University.

[9] Cremona, J. E. (2019). *Algorithms for Modular Elliptic Curves* (2nd ed.). Cambridge University Press.

[10] PARI/GP (2023). Version 2.15.4. https://pari.math.u-bordeaux.fr/.

[11] Johansson, F., et al. (2023). *mpmath: a Python library for arbitrary-precision floating-point arithmetic* (version 1.3.0). https://mpmath.org/.